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Palaeomagnetic dating and tectonic significance of dolerite intrusions in the Albany Mobile Belt, Western Australia

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Abstract

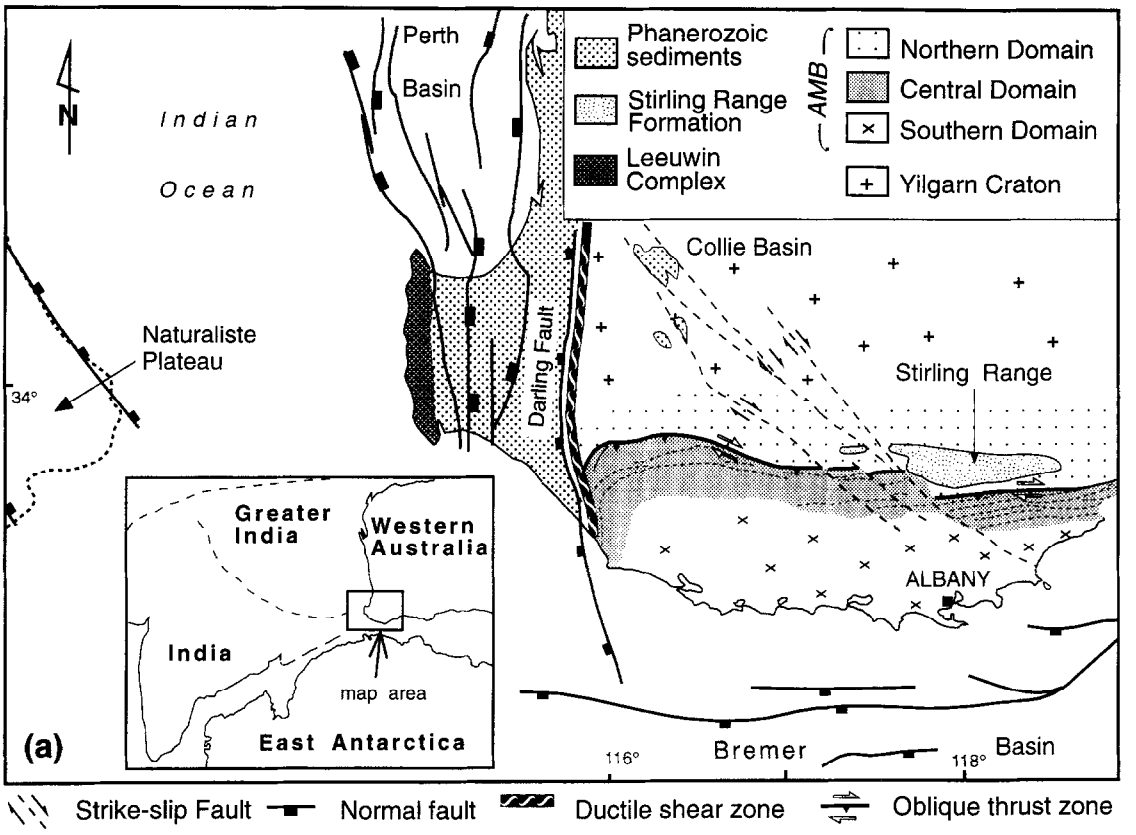
Dolerite dykes and sills in the Mesoproterozoic Albany Mobile Belt and Neoproterozoic Stirling Range Formation metasediments in southwestern Australia provide evidence for extensional events between Australia, East Antarctica and Greater India. Paleomagnetic analysis shows that primary remanence was carried by single domain magnetite for samples from seven localities. Palaeomagnetic poles for the majority of sites fall close to the earliest Cambrian part of the Australian apparent polar wander path (APWP). The interpreted earliest Cambrian dykes intrude both extensional (dominant) and shear fractures, indicating NNE–SSW extension between the Western Australian and East Antarctic Shields and a horizontal, ESE–WNW maximum compressive stress. Dyke emplacement appears to be contemporaneous with mafic volcanism in central and northern Australia and igneous intrusion in once contiguous parts of the East Antarctic Shield. These features are interpreted as being related to Gondwanaland assembly, and may represent a response to far-field stresses resulting from the collision between East and West Gondwanaland.

A second suite of NW-striking, mid-Carboniferous dykes is related to NE–SW extension and rifting between Greater India and the Western Australian Shield during early stages in the formation of the Perth Basin in a sinistral transtensional regime along the southern Western Australian margin. A palaeopole from the primary remanence in a NNW-striking dolerite dyke falls on the poorly defined Triassic to Early Cretaceous part of the Australian APWP, indicating a possible third phase of dyke emplacement. Regional tectonic constraints for rifting in the Perth Basin suggest that this youngest dyke most likely intruded during the Triassic.

1. Introduction

The Albany Mobile Belt on the southern margin of the Archaean Yilgarn Craton (Fig. 1a) in Western Australia is a Mesoproterozoic gneiss and granitoid complex [1,2] crosscut by several generations of ductile shear zones, faults and dolerite dykes. The generally E–W structural trends of the belt have been rotated into a N–S orientation towards its western margin by sinistral, Neoproterozoic ductile shear zones of the Darling Mobile Belt [3,4]. Neoproterozoic (Ediacaran) low-grade metasedimentary rocks of the Stirling Range Formation on the northern margin of the Albany Mobile Belt were deposited unconformably upon southern Yilgarn Craton and Mesoproterozoic basement [5] and deformed during two ?Early Paleozoic events [6]. The Darling Fault defines the western Yilgarn and Albany Mobile Belt margins, separating Archaean and

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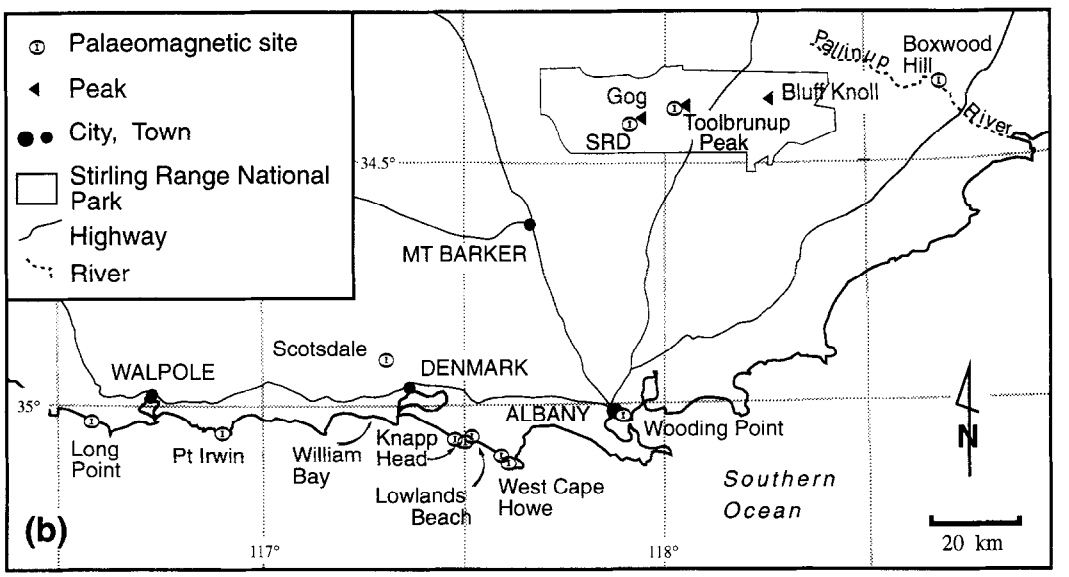


Fig. 1. (a) Simplified tectonic map of southwestern Western Australia showing the main tectonic subdivision of the Albany Mobile Belt (*AMB*) and adjacent terranes. Inset shows position within the Gondwanaland reconstruction [7]. (b) Location map of sample sites.

Proterozoic gneisses from the Phanerozoic Perth Basin. In Gondwanaland reconstructions (e.g., [7]) the Albany Mobile Belt is situated between the Yilgarn Craton and the East Antarctic Shield (Fig. 1a), linking with the Central Indian Tectonic Zone [1]. Southwestern Western Australia is therefore a key region for recording displacements between these terranes during Gondwanaland assembly [8] and breakup [9,10].

This paper presents the results of a palaeomagnetic and structural study of dolerite intrusions post-dating Mesoproterozoic tectonism

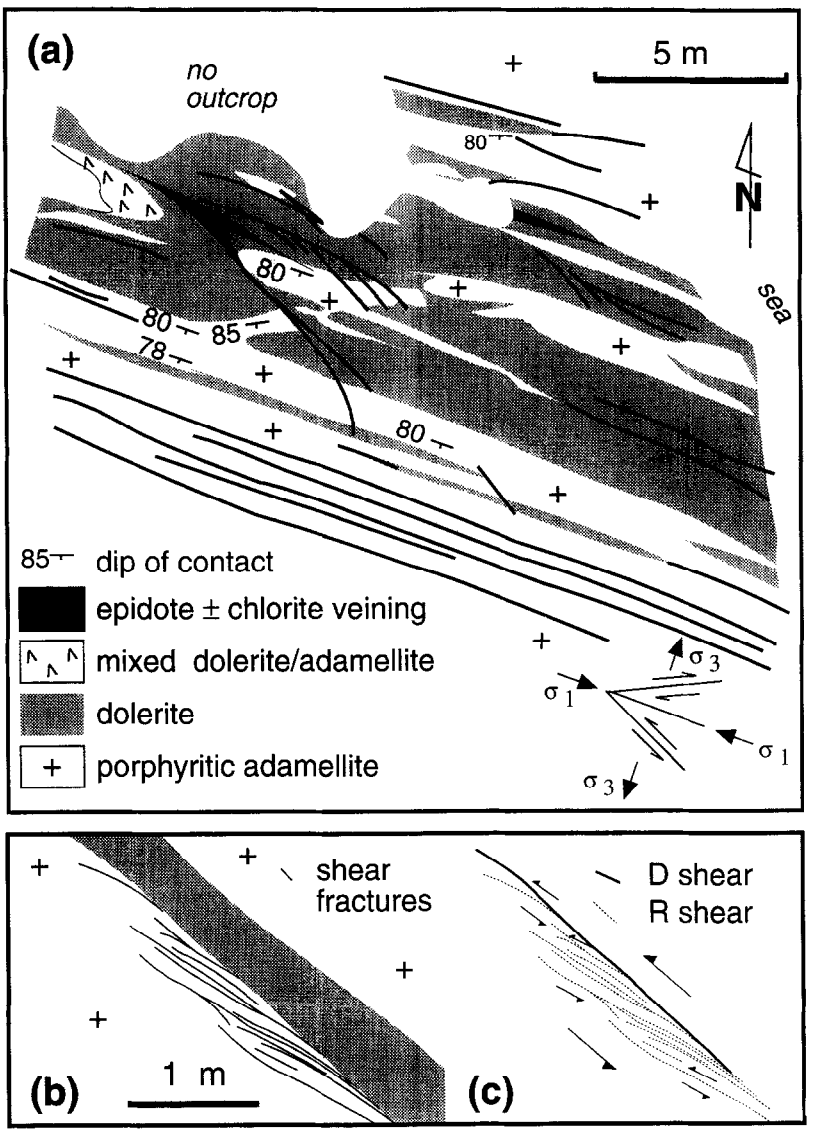


Fig. 2. Structural controls on Cambrian dyke emplacement. (a) Dykes at Lowlands Beach (locality LBD01) intruding extension fractures in quartz monzonite and cut by epidote-veined cataclastic shear zones. The suggested stress regime is indicated (principal stresses = $\sigma_1 \geq \sigma_2 \geq \sigma_3$; $\sigma_{H\ max}$ and $\sigma_{H\ min}$ are the maximum and minimum horizontal stresses). (b) Dolerite dyke at Hortons Beach, West Cape Howe emplaced along a sinistral fault zone. (c) Interpretation of fracture array on margin of dyke shown in (b) in terms of a Riedel shear (R) array. In this model the dykes have intruded a sinistral principal displacement shear (D).

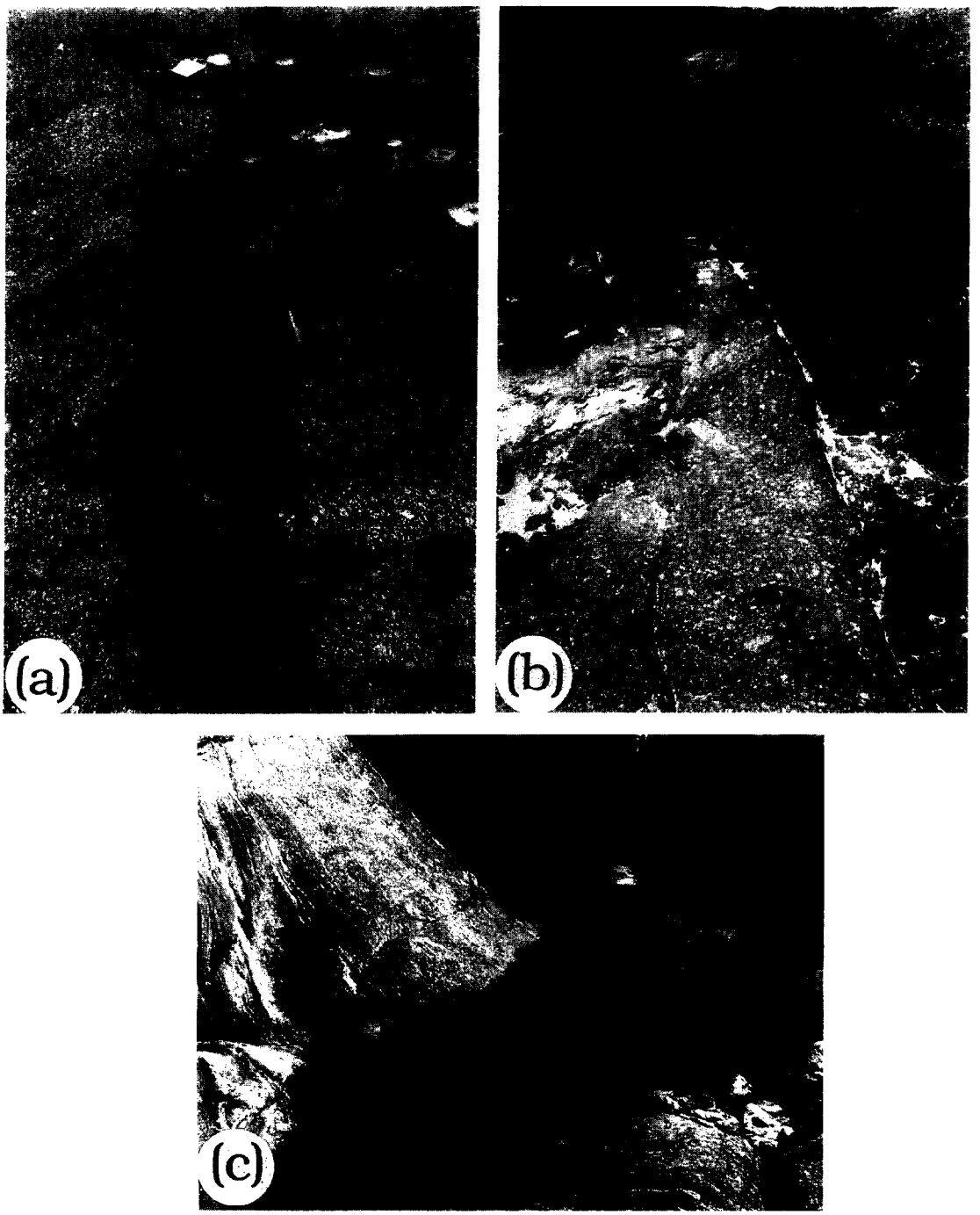


Fig. 3. Examples of structural controls on dyke intrusion. (a) En-echelon stepping of dolerite dykes defining a SE-striking, sinistral, brittle–ductile shear zone. Note link structure between two extensional fractures (Lowlands Beach, LBD05). (b) Multiple dyke injection (Lowlands Beach, LBD02–4). (c) NW-striking Carboniferous dyke intruding Mesoproterozoic gneiss (Long Point, LPD).

within the Albany Mobile Belt. Previously, the ages, and hence tectonic significance, of dolerite sills and ESE- and SE-striking dolerite dykes in the region were unknown; their presence has either been ignored, or related to Cretaceous to Tertiary breakup with Antarctica along the southern Australian margin. Problems in determining the emplacement age of dolerite dykes have been highlighted in geochronological studies [e.g., [11]] which show that, unless use of the U-Pb method (e.g., on baddeleyite) is possible, other radiometric methods such as Ar/Ar and K/Ar are often fraught with problems and may provide imprecise ages and ages that are younger than that of the dolerite emplacement. In the quartz tholeiite dolerite dykes of the Albany Mobile Belt no baddeleyite has been recognised, precluding the use of the U/Pb method. In K/Ar and Ar/Ar whole-rock dating there is also a likelihood of Ar loss due to possible ~ 500 Ma (and possibly younger) low-temperature thermal pulse(s) (as described for contiguous Antarctic terranes [12]) which may mean that minimum ages would only be derived using these methods. The palaeomagnetic method has been used here to provide a non-radiometric method for estimating the age of dolerite emplacement.

2. Description of sample localities

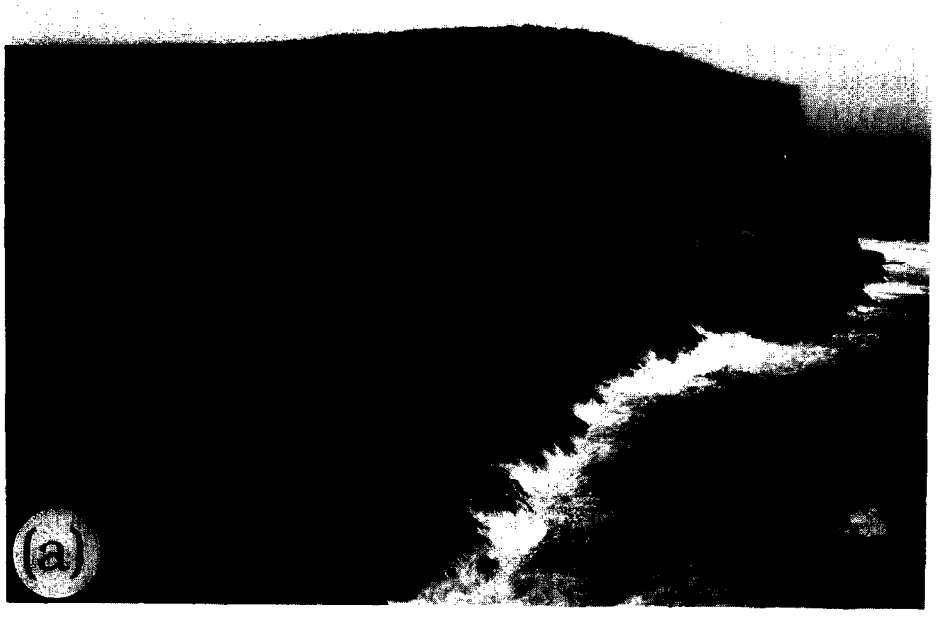
Ninety-eight samples (diamond cores and oriented blocks) were collected from fourteen dolerite intrusions in both Mesoproterozoic granitoids and gneisses and the Stirling Range Formation (all place names and sample localities described below are shown in Fig. 1b). Both magnetic, and in the majority of cases, sun compass readings were used in orienting samples. Localities were chosen up to 80 km from each other in a N–S orientation in order to investigate the possibility for tilting after dolerite emplacement, such as during breakup with Antarctica [9].

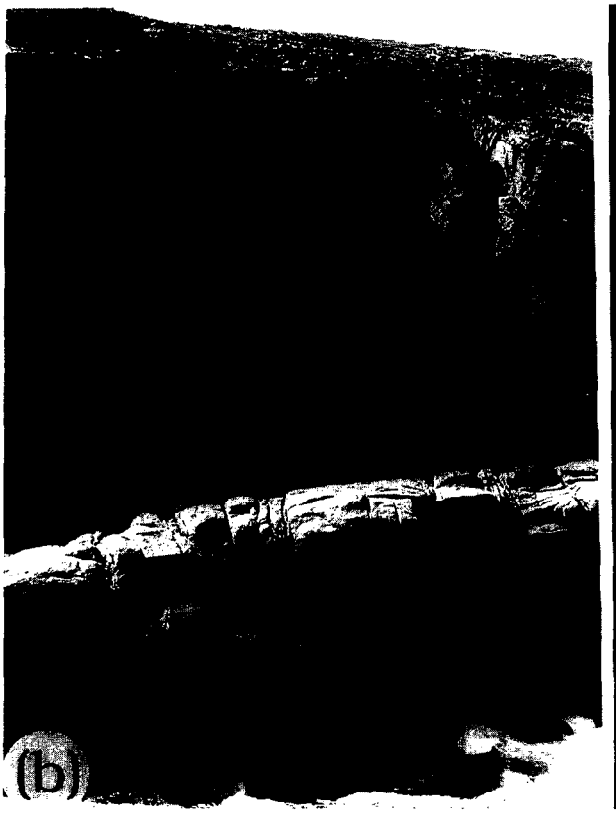
2.1. Albany Mobile Belt sites

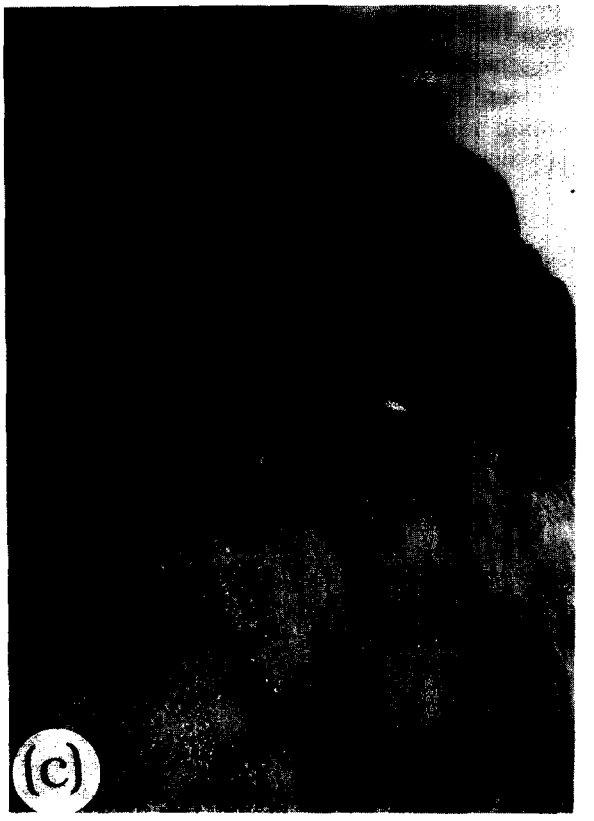
Five highly magnetic sulphide-bearing dolerite dykes (several centimetres to 5 m in width) in quartz monzonite were sampled at Lowlands Beach (LBD). Although there has been assimilation and mixing of fragments of host rock in one dyke, the margins are generally sharp, often with extensive epidote and chlorite alteration. The dykes intrude epidote ± chlorite-filled extension fractures striking 095–110° (Fig. 2a). En-échelon stepping of some dykes (Fig. 3a) suggests intrusion along incipient strike-slip fault zones, and some dykes are cut by NW-striking sinistral shear fractures. Samples LBD02–LBD04 were drilled from a closely spaced group of 30–90 cm wide dolerite dykes that strike 280–295° in an area of multiple magma injection along a zone of fracturing in the hostrock. Narrow screens of highly metasomatised (K-feldspar + epidote alteration) quartz monzonite controlled by ENE- and SE-striking fractures separate some dykes for part of or along the entire dyke length (e.g., Fig. 3b).

Block samples were taken at varying distances from the margin of a broad, irregular dolerite body cutting Mesoproterozoic gneiss and amphibolites near Boxwood Hill (BH) within the Northern Domain of the Albany Mobile Belt. Contacts with country rocks are sharp, and are controlled by fractures that strike at 350° and that are either subvertical (115°) or steeply dipping to the east.

The ~ 100 m thick olivine tholeiite and quartz dolerite sill at West Cape Howe (Fig. 4a) has been interpreted from a structural and ground magnetic study as a 'stepped intrusion', steepening locally at depth and continuing with flatter dips again at a deeper level [13]. Sill margins are controlled by gently NE- and SW-dipping (Figs. 4b, 5a and 5b) and steep fracture arrays (Fig. 4c) in the host granitoid. Aerial views show that the gently dipping fractures are accompanied by extensive flat-lying pegmatite sheets. The angular relationships between the fractures and the pegmatites, together with field observations of the displacement of Mesoproterozoic pegmatites and gneissic phases, indicates that these structures represent thrust faults/shear joints and extension fractures formed during subhorizontal NE–SW compression (note that dolerite intrusion into pre-existing thrust faults is seen at other localities, for example at William Bay where steps on the slickenside margins of a ~ 1.5 m thick do-


(a)


(b)


(c)

lerite sheet dip 40° SW and indicate thrust faulting prior to dolerite intrusion). Vertical, NW-striking dolerite dykes (apparent on aerial photographs of southern cliff faces) suggest that steeply dipping structures may have served as feeder dykes, with migration of magma along pre-existing gently dipping fractures forming the sill. The sill and hostrock are cut by conjugate, brittle, transcurrent shear zones and extension fractures (Figs. 4a, 4c, 5c and 5d) containing epidote ± chlorite assemblages. The displacement of feldspars along microshears, micro-Riedel shear arrays and en-échelon epidote-filled extensional fractures in thin sections from cataclastic and ultracataclastic shear zones indicates ENE dextral and SE sinistral displacements. Evidence for sinistral displacement pre-dating dyke intrusion also comes from en-échelon Riedel shears (Figs. 2b and c) on the margin of a ~ 1 m wide 126°-striking dolerite dyke intruding a brittle–ductile shear zone. A drillcore was taken from coarse-grained dolerite within the main body of the sill (site WC01), and from the contact between the dolerite and a slice of wallrock incorporated in the intrusion (the raft in Figs. 4a and c; site WC02) where a chilled margin is developed at the contact with the adamellite.

A steeply dipping dyke (sample sites SD01–3) near Scotsdale (SD) lies on strike with scattered outcrops of the West Cape Howe sill, suggesting that either both bodies may be part of the same intrusion (continuous for > 40 km) or that dolerite intrusions in both areas were controlled by the same structure. Although contacts with the gneissic country rock are rare due to the lack of outcrop, a 110° orientation of the margin (of finer grain size than the rest of the body examined) was recorded at one locality. No major magnetic feature is discernible on regional aeromagnetic data. It is therefore possible that these scattered dolerite outcrops represent a series of intrusions dominantly controlled by ENE extensional fractures that step along a NW-striking fault zone (in a manner similar to dykes stepping on the outcrop scale at Lowlands Beach, Fig. 3a). Samples from Wooding Point, Albany differ in composition from other dykes in that they are of variable grain size and composition, are extensively chloritised and carbonate-altered, and contain a high percentage of country rock xenoliths and carbonate pods. In thin section, multiple stages of magma injection are indicated by fragments of coarse-grained dolerite in a finer dolerite matrix. There is a large percentage of highly deformed, strongly resorbed quartz xenoliths, and a high titanite and magnetite component. These dykes have been incorrectly described [14] as 'clastic dykes'; the abundance of microphenocrysts of plagioclase, and remnant textures of amphiboles and/or pyroxenes replaced by chlorite, attest to a mafic intrusive origin. Fractures synchronous with dyke intrusion lie within a 130°-striking, sinistral fault zone (a prominent Landsat lineament) and cut conjugate strike-slip shear fractures bisected by NE-striking quartz-filled extension gashes. All structures are cut and displaced by a third set of barren, NNE- and NNW-striking conjugate shear fractures.

A second set of dolerite dykes is characterised by freshness and the absence of epidote veining and country rock alteration. Block samples were taken from a 1–1.5 m wide fine-grained dolerite (Fig. 3c) striking 330° and intruding ortho- and paragneisses east of Long Point (LPD), and from a 1.3 m wide dolerite dyke striking 165° and dipping at ~ 60° WSW cutting migmatitic gneisses northwest of Point Irwin. Core samples were taken from a dolerite dyke striking between 145° and 160° on the western side of Knapp Head (KHD).

Fig. 4. The West Cape Howe sill. (a) View of the raft sample site, a granitic xenolith (light) within the dolerite sill. Height of cliff section is 80 m. Samples were drilled on the upper margin of the xenolith, as shown in (c). (b) Structural control of gently NE- and SW-dipping fractures (formed during a NE–SW shortening event prior to dolerite intrusion) on earliest Cambrian sill emplacement (helicopter photograph). (c) The raft sample site. Note steep cataclastic shear zones in both country rock and, to a lesser extent, dolerite. These structures, which are contemporaneous with mafic igneous intrusion in the Albany Mobile Belt, have also exerted a structural control on the sill emplacement.

2.2. Stirling Range dykes

Aeromagnetic imagery over the Stirling Range Formation shows ESE-striking dolerite dykes and ENE-striking dextral faults [6]. Samples of coarse-grained fractionated dolerite were drilled from a dyke striking 130° that intrudes siltstones southwest of Mt. Gog (site SRD). A ~9 m wide dolerite dyke on the southwestern spur of Toolbrunup Peak (sample TD) contains a penetrative (igneous?) foliation striking 155° (i.e., parallel to the fracture-controlled margins) cut by a second set of closely spaced steeply W-dipping fractures striking 020°. From aerial photographs, fractures

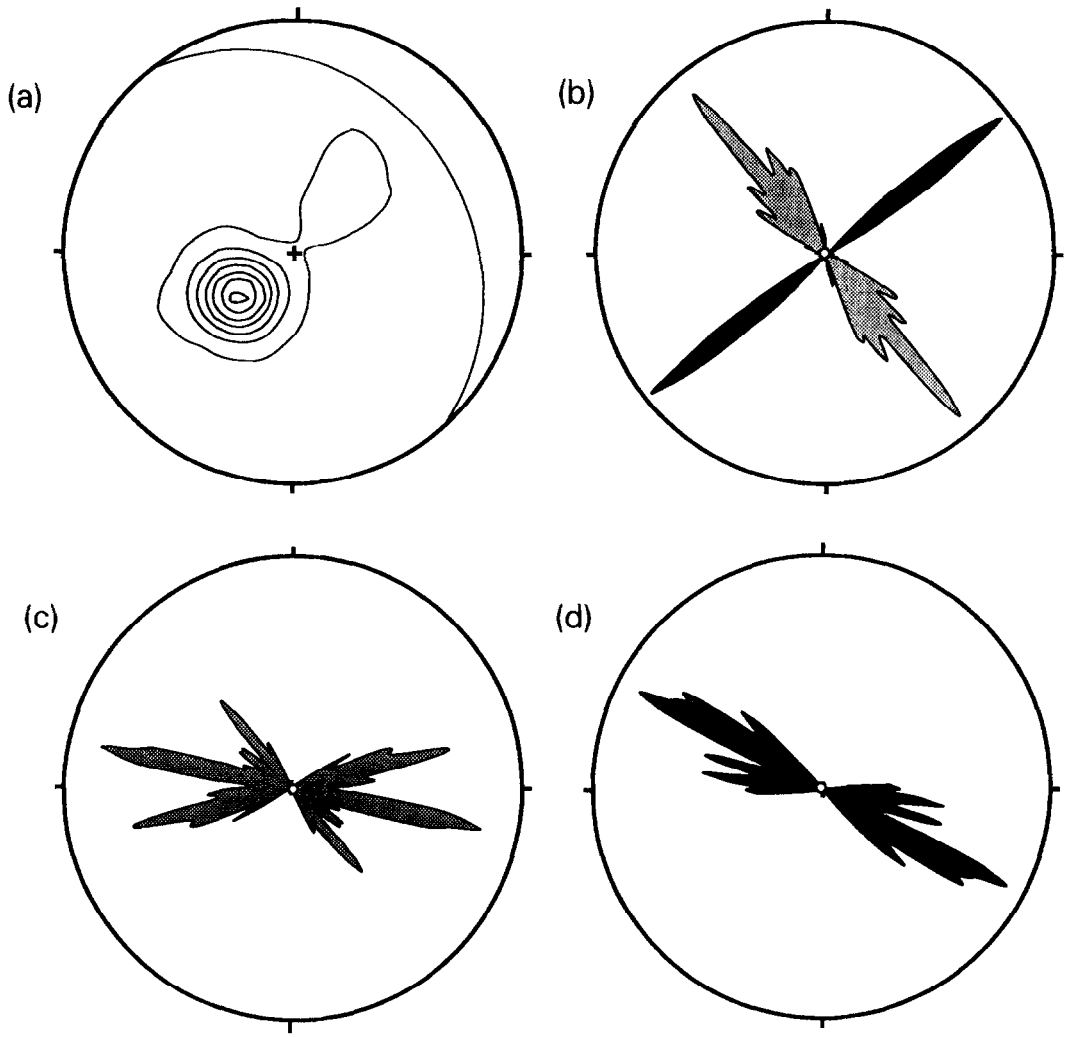


Fig. 5. Stereographic projections (lower hemisphere, equal area) of structural data from West Cape Howe. (a) Contours of poles to fractures developed during NE–SW compression controlling sill emplacement (as shown in Fig. 4b). (b) Rose diagram of data shown in (a) in grey (84 data, 5° intervals, radius = 22%) with additional, vertical pegmatites (black, 8 data, radius = 50%) striking parallel to the implied NE–SW compression direction. (c) Rose diagram of fractures and cataclastic shear zones in adamellite, central West Cape Howe area. Maxima correspond to ENE dextral and SE sinistral shear zones, which are bisected by extension fractures (43 data, radius = 15%). Fractures correspond to those formed during the earliest Cambrian event controlling dolerite emplacement. (d) Rose diagram of fractures and cataclastic shear zones in adamellite, eastern West Cape Howe area, along ESE-striking sinistral cataclastic shear zones (20 data, radius 20%).

with the same strike as the dyke are seen to be common and to post-date folding. A dolerite dyke near the base of Bluff Knoll has, however, been deformed and metamorphosed along with host metasediments, indicating the likelihood of variability in the timing of dolerite intrusion with respect to deformation of the Stirling Range Formation.

3. Palaeomagnetic analysis

3.1. Methods

One or two specimens from each sample were subjected to stepwise demagnetisation in the University of Western Australia's palaeomagnetism laboratory. Thermal demagnetisation was carried out using a Schonstedt TSD-1 demagnetiser and alternating field (AF) demagnetisation was achieved using a Molspin demagnetiser. Magnetic remanence was measured on a computer-driven Digico spinner magnetometer installed in a space of low magnetic field. Data analysis was accomplished by employing the principal component analysis technique [15] in conjunction with orthogonal plots [16] using the ROCMAG software package developed by the CSIRO rock magnetism laboratory.

Out of the total of fourteen intrusive bodies studied, four failed to retain their primary remanence: Dyke PI failed to retain any consistent remanence, most likely due to a lightning strike, and dyke LBD01 retained a viscous present field overprint only. Samples from the Wooding Point gave no consistent remanent directions probably due to the block rotations during the multiple magma injections. The three samples from the dyke at the Toolbrunup Peak also failed to give a consistent direction. The remainder retain either a high-temperature component (HTC) (with an unblocking temperature of $\sim 575°$ C) only, or both high- and low-temperature components (LTC) (with an unblocking temperature of $\sim 300°$ C). Representative orthogonal plots are given in Figs. 6 and 7 and show that magnetite is the prime carrier of the magnetic remanence. A summary of the magnetic results is given in Table 1.

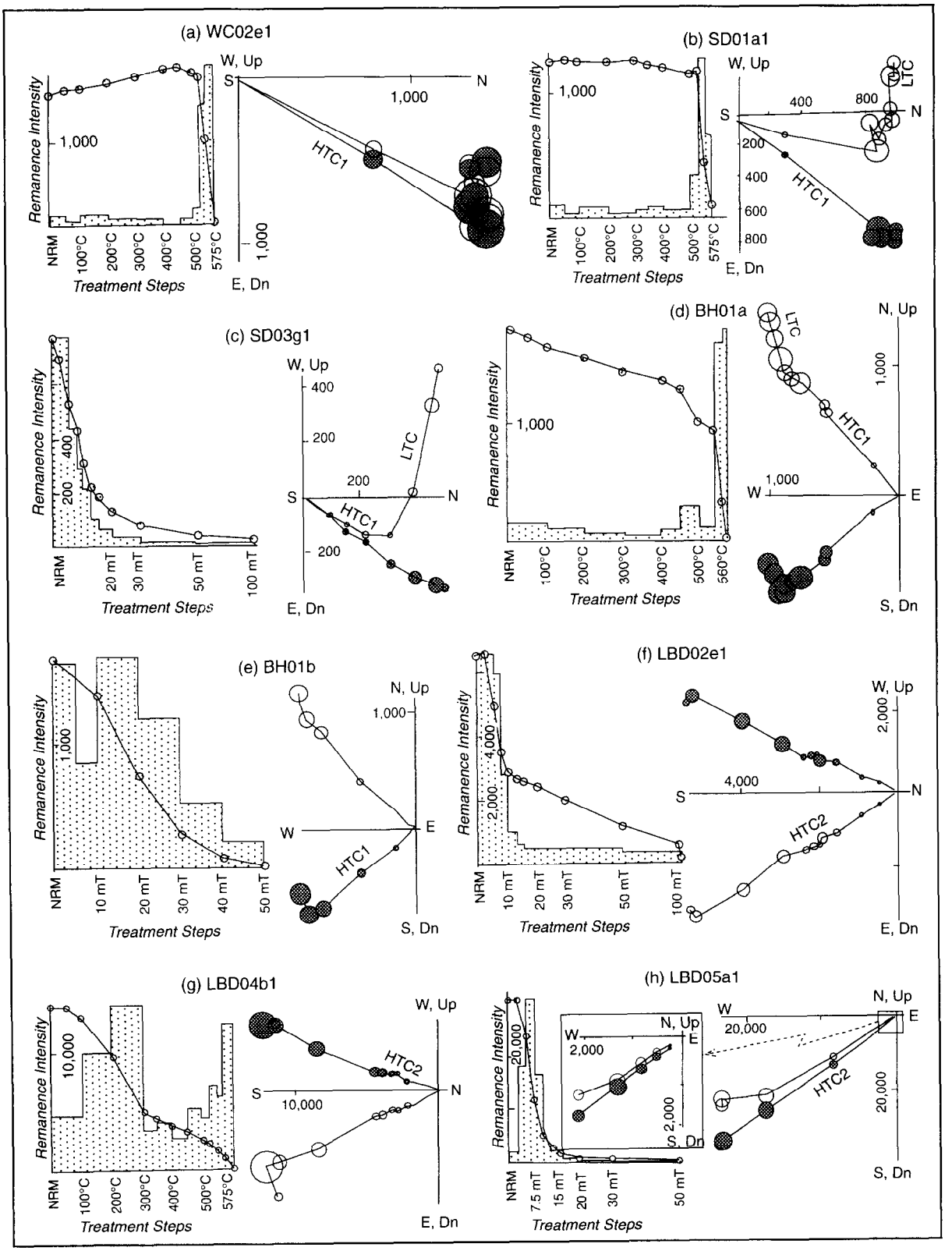
3.2. High-temperature component and ages of dolerite intrusion

The directions of the HTC fall into four groups: (1) HTC1 (sites WC01–02, SD01–03 and BH) directed either shallowly downward to the northeast (Figs. 6a–c and 8a–e), or upwards to the southwest (Figs. 6d, 6e and 8f); (2) HTC2 (sites LBD02–05) directed shallowly downward to the south-southwest (Figs. 6f–h and 8g–j); (3) HTC3 (sites LPD and SRD) directed moderately to steeply downward (Figs. 7a, 7b, 9a and 9b); and (4) HTC4 (site KHD) directed steeply upward to the northwest (Figs. 7c, 7d and 9c).

All four groups are considered to be of primary origin because (1) the thermal stability of their remanence would require a thermal event with a temperature close to the Curie temperature of single domain magnetite (i.e., $\sim 580°$ C) to remagnetise the remanence [17], and there is no evidence for such an event in the region since the Neoproterozoic, (2) the four high-temperature remanent components coexist in the same region, which is evidence against the likelihood of such an overprinting event, and (3) the magnetite in the intrusive bodies is of primary origin (based on thin section observations).

Taking all the HTCs as primary remanence, to date these dolerite bodies palaeomagnetically the positions of their corresponding palaeopoles on the known apparent polar wander path (APWP) for Australia were examined. In Fig. 10, the latest Proterozoic to Permian part of the path is defined using A- and B-class palaeomagnetic poles [18]. The latest Proterozoic to Cambrian segment of the path is based on data from Australia and India [18], since the African and South American blocks may not have joined with East Gondwanaland (Australia, India and East Antarctica) at that time [18–22]. The Ordovician to Permian segment is defined using poles from all the major Gondwanan continents [23,24]. The Mesozoic segment is adopted from [25]. Note that the APWP for latest Proterozoic to Early Cambrian, Devonian and Permo-Carboniferous intervals is well defined by high-quality palaeomagnetic poles.

Paleopoles calculated according to the four groups of HTC are plotted in Fig. 10 together


(a) WC02e1
Remanence Intensity
1,000
NRM
100°C
200°C
300°C
400°C
500°C
575°C
Treatment Steps
W, Up
S
N
1,000
HTC1
1,000
E, Dn
(b) SD01a1
Remanence Intensity
1,000
NRM
100°C
200°C
300°C
400°C
500°C
575°C
Treatment Steps
W, Up
LTC
400
800
S
N
200
400
600
800
HTC1
E, Dn
(c) SD03g1
Remanence Intensity
200
400
NRM
20 mT
30 mT
50 mT
100 mT
Treatment Steps
W, Up
400
200
LTC
200
S
N
HTC1
200
E, Dn
(d) BH01a
Remanence Intensity
1,000
NRM
100°C
200°C
300°C
400°C
500°C
560°C
Treatment Steps
N, Up
LTC
HTC1
1,000
1,000
W
E
S, Dn
(e) BH01b
Remanence Intensity
1,000
NRM
10 mT
20 mT
30 mT
40 mT
50 mT
Treatment Steps
N, Up
1,000
W
E
HTC1
S, Dn
(f) LBD02e1
Remanence Intensity
4,000
2,000
NRM
10 mT
20 mT
30 mT
50 mT
100 mT
Treatment Steps
W, Up
2,000
4,000
S
N
HTC2
E, Dn
(g) LBD04b1
Remanence Intensity
10,000
NRM
100°C
200°C
300°C
400°C
500°C
575°C
Treatment Steps
W, Up
HTC2
S
N
10,000
E, Dn
(h) LBD05a1
Remanence Intensity
20,000
NRM
7.5 mT
15 mT
20 mT
30 mT
50 mT
Treatment Steps
N, Up
W
E
2,000
2,000
S, Dn
W
20,000
N, Up
E
HTC2
20,000
S, Dn

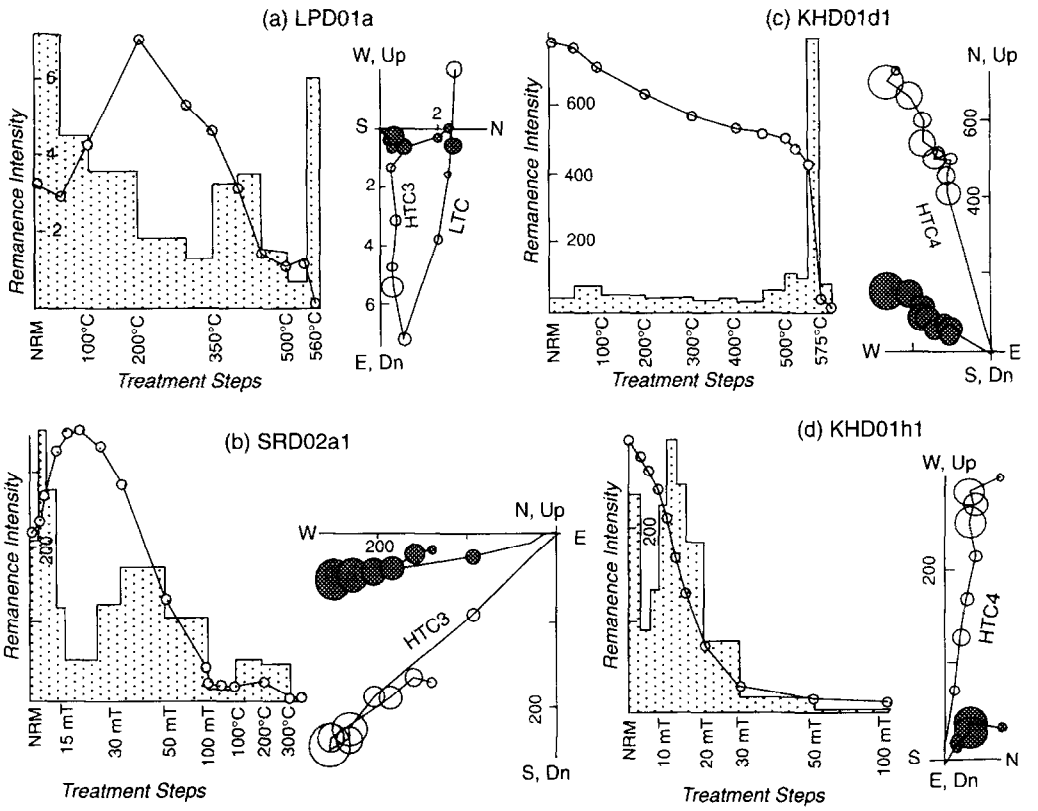


Fig. 7. Representative orthogonal plots and intensity curves of samples from Long Point dyke (LPD), Stirling Range Drive dyke (SRD) and Knapp Head dyke (KHD). Conventions are the same as in Fig. 6.

with their 95% confidence error ellipses. Before interpreting the age of these intrusions it must be borne in mind that each of the intrusive bodies studied recorded the direction of the instantaneous palaeomagnetic field only. The position of its corresponding palaeomagnetic pole is, therefore, subject to errors of over 10° due to the secular variation of Earth's magnetic field (the present magnetic South Pole is ~ 25° away from the geographic South Pole). The only way to reduce this error is to calculate a time-averaged palaeomagnetic pole.

In Fig. 10, as the poles of HTC1 (WC, SD and BH) fall close to the earliest Cambrian segment of the APWP, and significantly away from both older [26] and younger segments, they are regarded as of Early Cambrian age. The positions of these three poles coincide with the palaeopole (pole AP in Fig. 10) obtained from the earliest Cambrian(?) Antrim Plateau Volcanics of northern Australia [27], although this pole is not classified as a key pole due to the lack of detailed demagnetisation.

The four LBD poles calculated from the HTC2

Fig. 6. Representative orthogonal plots and intensity curves for samples from West Cape Howe sill (WC), Scotsdale dyke (SD), Boxwood Hill dyke (BH) and Lowlands Beach dykes (LBD) with HTC1 (a–e) or HTC2 (f–h). The unit for remanence intensity is 1×10^{-6} emu cm^{-3} (mA/m). Small circles in the orthogonal plots are standard errors for each measurement step, where the shaded circles are projections on the horizontal plane, and the open circles are projections on the vertical plane.

Table 1
Summary of the palaeomagnetic results

Site	Location		Sample Number	Remanent Direction				South Pole Position			
Name	(°S,	°E)	(n2/n1/no)#	Dec. (°)	Inc. (°)	α_{95}(°)	k	Plat.(°S)	Plong.(°E)	dp(°)	dm(°)
HTC1											
WC01	35.13	117.61	3/3/3	34.1	29.3	12.9	91.7	29.8	336.1	7.9	14.2
WC02	35.13	117.61	3/3/3	352.0	22.2	11.6	114.9	27.8	335.6	6.5	12.3
Mean of 2 (pole WC)								28.8	335.8		
SD01	34.93	117.27	5/5/6	40.7	20.3	7.0	120.3	30.5	345.3	3.8	7.3
SD02	34.93	117.27	3/3/4	43.8	24.1	14.9	69.2	26.9	346.5	8.5	15.9
SD03	34.93	117.27	5/5/8	43.5	30.0	10.3	56.5	24.4	343.8	6.3	11.4
Mean of 3 (pole SD)								27.3	345.2	A_{95}=5.0	
BH01	34.38	118.70	5/5/6	233.0	-41.6	4.4	307.6	13.0	347.2	3.3	5.4
HTC2											
LBD02	35.08	117.51	6/6/6	198.9	34.5	4.4	235.8	66.8	348.5	2.9	5.0
								(pole LBD2)			
LBD03	35.08	117.51	6/6/6	209.1	20.7	10.0	45.9	54.0	352.0	5.5	10.5
								(pole LBD3)			
LBD04	35.08	117.51	5/5/5	204.5	23.5	6.8	125.8	58.2	347.7	3.9	7.2
								(pole LBD4)			
LBD05	35.08	117.51	3/7/7	227.8	27.9	31.9	16.0	42.7	014.6	19.1	34.9
								(pole LBD5)			
Mean of 4								56.0	357.7	A_{95}=14.5	
Mean of all the HTC1 and HTC2 poles (pole M): N=10, (37.8°S, 346.9°E) A_{95}=11.7°, K=18.1											
HTC3											
LPD	35.03	116.58	5/5/5	318.8	81.1	12.4	38.9	21.5	104.2	23.1	24.0
SRD	34.42	117.93	4/4/10	231.3	70.4	25.0	14.4	49.4	074.0	37.4	43.2
HTC4											
KHD	35.09	117.47	7/8/9	290.0	-71.8	5.0	145.7	39.4	159.1	7.7	8.8
LTC											
WC01*	35.13	117.61	3/3/3	352.0	22.2	11.6	114.9	42.7	286.9	6.5	12.3
WC02*	35.13	117.61	9/9/10	299.6	-50.8	10.2	26.6	40.2	193.6	9.3	13.8
SD01	34.93	117.27	5/5/6	42.0	-75.3	14.4	29.1	52.2	086.8	24.1	26.4
SD03	34.93	117.27	7/8/8	37.1	-68.3	7.0	75.0	58.8	070.9	9.9	11.8
LBD01	35.08	117.51	8/8/12	338.5	-54.7	10.4	29.4	72.5	200.8	10.4	14.7
LPD	35.03	116.58	4/4/5	24.8	-72.8	7.9	136.4	61.6	269.0	12.5	14.1

n_0 = total number of samples; n_1 = number of samples which retained a stable remanence; n_2 = number of samples used for calculating the site mean direction. * Remanence regarded as erroneous.

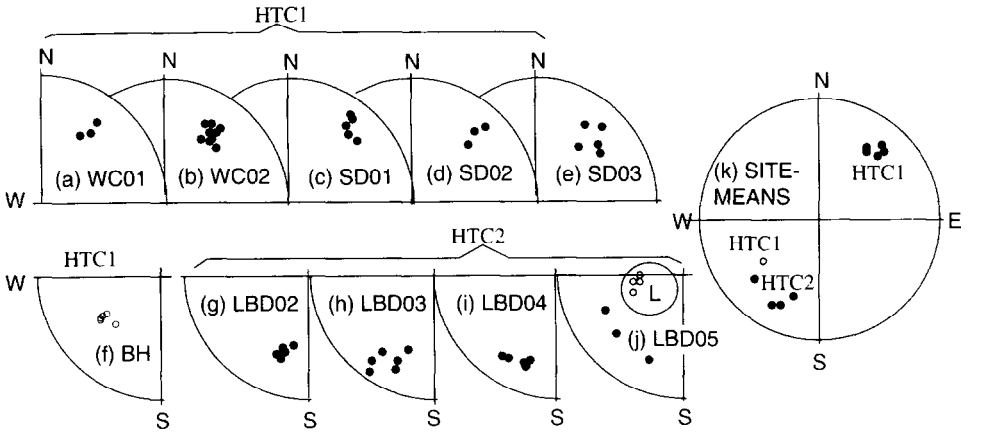


Fig. 8. Equal area projections of high-temperature component directions (HTC1, a–f; HTC2, g–j) of individual sites (a–j) and site means (k). • = Lower hemisphere projections; ○ = upper hemisphere projections. The four directions marked with *L* in (j) are considered to be remanent directions due to a lightening strike, as demonstrated by their enormous remanence intensity despite having the same magnetic susceptibility as the other samples from that site.

of four adjacent dykes at Lowlands Beach fall between the Early Cambrian and Late Devonian segments of the APWP, and overlap with the poorly known Silurian segment (Fig. 10). Judging from their positions alone, their age could be either earliest Cambrian or Silurian to Devonian. However, as discussed above, the remanence from the four dykes, which were emplaced adjacent to each other (Figs. 3a and b) and share the same magnetic polarity, probably represents a record of an instantaneous palaeomagnetic field, indicating that their corresponding palaeopoles should not be used alone for age determination. Furthermore, based on structural analysis these four WNW-trending dykes and the sill with the HTC1 remanence were probably controlled by a single tectonic event. An overall mean (M) of both the HTC1 and HTC2 poles falls on the earliest Cambrian APWP (Fig. 10). The LBD dykes are thus interpreted to be also of earliest Cambrian age. Note that the mixed polarities within these seven poles (i.e., WC and SD normal, BH and LBD02–5

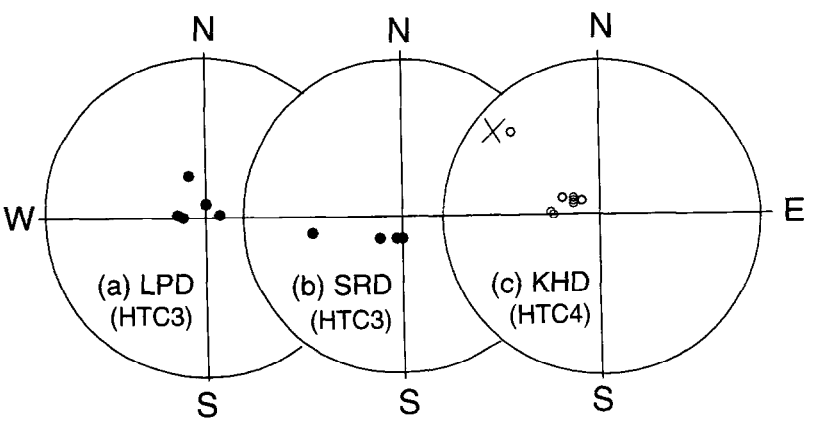


Fig. 9. Equal area projections of high-temperature component directions (HTC3, a and b; HTC4, c) of dykes LPD, SRD and KHD. Conventions are the same as in Fig. 8. The direction marked with a cross in (c) is taken as an outlier.

reversed) should have averaged out the secular variation error. The large scatter between the pole positions could partly be due to secular variation and partly due to minor block rotations during the breakup of Gondwanaland.

Poles LPD and SRD fall close to the mid-Carboniferous part of the APWP. The reversed polarity indicates that they could have intruded during the Permo-Carboniferous Kiaman reversed magnetic interval [28]. Pole KHD falls on the poorly defined Triassic to mid-Cretaceous segment of the APWP, possibly indicating the youngest intrusive event in the region.

3.3. Low-temperature components

Low-temperature components (LTCs) have been revealed from six sites in four intrusive bodies. Apart from two sites from the West Cape Howe sill, the directions of the remaining four sites all fall close to the present geomagnetic field direction (Fig. 11), indicating a recent viscous

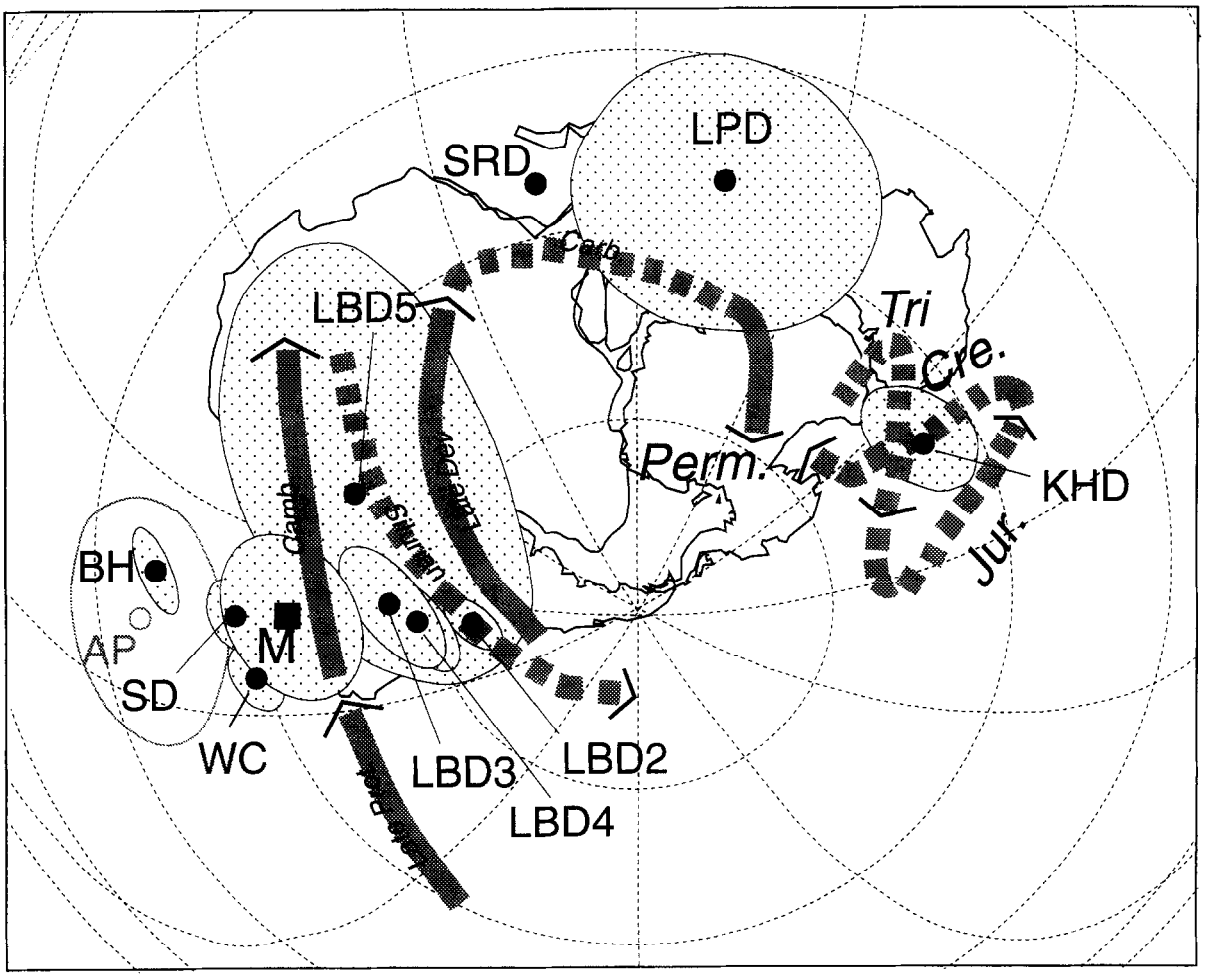


Fig. 10. Latest Proterozoic to mid-Mesozoic apparent polar wander path (APWP) for Australia, and positions of the palaeomagnetic poles mentioned in text. Continuous parts of the APWP are better known than the dashed parts. The 95% confidence ellipses for most of the poles are also shown. All the poles obtained from this study (solid symbols) are listed in Table 1.

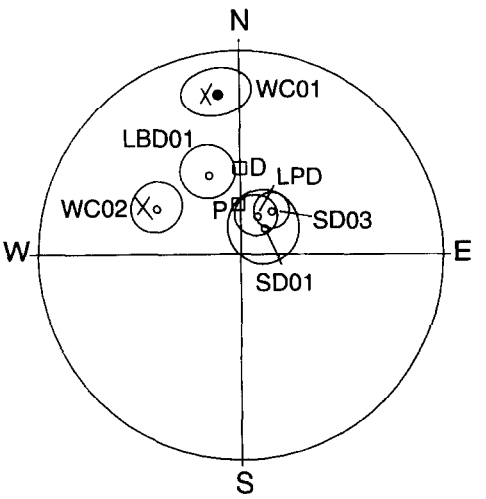


Fig. 11. Equal area projections of the low-temperature component (LTC) from dykes/sill WC, SD, LBD01 and LPD. Small circles are the 95% confidence circles of each site mean. D and P = the directions of the present dipole field and present geomagnetic field respectively. Conventions are the same as in Figs. 8 and 9.

origin. The two sites from the West Cape Howe sill are regarded as abnormal because of the lack of agreement between them.

4. Structural controls on dolerite intrusion

4.1. Application of dyke orientations in determining the palaeostress field

Before interpreting the tectonic significance of dolerite emplacement in the Albany Mobile Belt it is necessary to review the accepted relationships between dykes and palaeostress fields and pre-existing structures. Dykes are generally thought to intrude perpendicular to the applied minimum principal stress σ_3 [29], although there are certain additional constraints on the process [30]. Magma intrusion along pre-existing vertical fractures depends on both the angle between the fractures and the minimum principal stress and on the relative values of magmatic pressure and regional principal stress [31], so that intrusion may also take place into pre-existing shear fractures [31,32] (as illustrated in Fig. 2b). Dolerite dyke intrusion is not limited to classical rift systems which result from regional extensional tectonism (i.e., where the maximum compressive stress σ_1 is vertical) and is common in cratonised terrains inboard from a convergent tectonic margin. In this second case σ_1 is horizontal (= maximum horizontal compressive stress $\sigma_{H\ max}$) and the dykes strike parallel to σ_1 [33]. Where $\sigma_1 = \sigma_{H\ max}$ the direction of dyke propagation is not significantly disturbed by pre-existing fractures, whereas in areas where σ_1 is vertical dyke intrusion may occur along pre-existing fractures with variable orientations [33]. A subhorizontal orientation of the maximum compressive stress does not, however, necessarily reflect a relationship with a collisional orogen. During rifting in Iceland, a subhorizontal orientation of maximum compressive stress both parallel and perpendicular to the spreading axis is common [34] and conjugate strike-slip faults bisected by extensional fractures commonly dominate over normal faults. Dolerite dykes in Iceland generally intrude extensional fractures parallel to the axis of rifting, although the dyke patterns (e.g., [35]) suggest that some dykes also intrude shear fractures (in the manner described by Tokarski [32]).

4.2. Structural controls on dykes in the Albany Mobile Belt

From the field descriptions outlined in Section 2 the earliest Cambrian dykes are seen to be controlled by extensional fractures that acutely bisect cataclastic transcurrent shear zones that indicate SSW–NNE extension/WNW–ESE compression (i.e., where $\sigma_1 = \sigma_{H\ max}$). NE- and SW-dipping thrusts pre-date, and in part control the intrusion of dolerite sheets. As dykes cut long-wavelength folds with NW–SE axial traces and conjugate shear fractures (N dextral and ENE sinistral) bisected by NE-striking extensional veins, at least one period of regional NE–SW shortening across the Albany Mobile Belt must have occurred prior to the event during which the dykes were emplaced, although the timing of this event remains uncertain. As the same structures associated with dyking are ubiquitous throughout

the Albany Mobile Belt and in neighbouring terranes, dyke emplacement was controlled by a regional stress field and not by locally induced stresses (such as above a magma chamber). The Albany Mobile Belt therefore provides an example in which dyke intrusion is not associated with normal faulting, as is the case in many rift systems (e.g., [34,35]), but shows structural relationships similar to those associated with continental collisional zones [33].

Fewer structural data are available for the Carboniferous dyke suite due to the limited number of dykes attributed to this event. The dykes are interpreted to have intruded extensional fractures within a NE–SW extensional deformation regime. The Carboniferous dyke in the Stirling Range (SRD) strikes parallel to basement structures interpreted from gravity data that controlled Neoproterozoic sedimentation, suggesting the reactivation of these structures.

5. Regional correlations and tectonic significance of Cambrian dolerite emplacement

Difficulties are faced in correlating dolerite intrusion in the Albany Mobile Belt with other areas because of inadequate data and uncertainties in relating stratigraphic and geochronometric ages for the Early Cambrian (as emphasised in Antarctic correlations [36]). However, correlations may be suggested on the basis of available

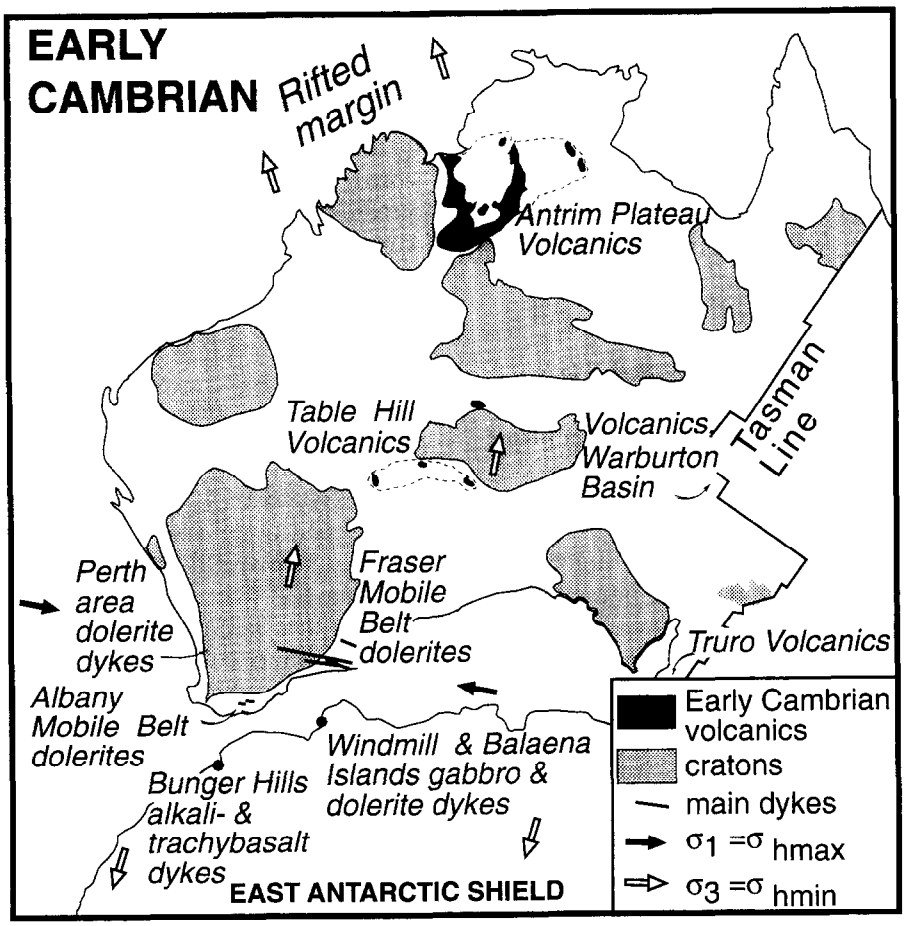


Fig. 12. Australian and contiguous East Antarctic terranes showing location of Early Cambrian volcanism and dolerite intrusions.

ages, orientations and relative timing with respect to regional deformation events.

Approximately E-W striking dolerite dykes within the southern Yilgarn Craton [37,38] have previously been thought to belong to either the Mesoproterozoic (1300 Ma?) Gnowangerup dyke swarm [39,40] (generally undeformed and unmetamorphosed, but deformed and recrystallised on the Albany Mobile Belt margin at 1190 Ma) or the 2.4 Ga [41] Widgemooltha dyke suite. Preliminary palaeomagnetic studies in the southwestern Yilgarn Craton show that one of the E–W dykes previously thought to be part of the Widgemooltha dyke suite could be Early Cambrian in age (Li and Trench, unpublished). It is, therefore, highly probable that dykes classified as the Gnowangerup dyke swarm represent several phases of intrusion along the southern Yilgarn Craton, including Cambrian dykes.

Aeromagnetic images of the southern Fraser Mobile Belt and Yilgarn Craton show dykes of the Widgemooltha dyke suite terminating at the mobile belt boundary [1,8]. SE-striking, highly magnetic basic dykes bisecting conjugate strike-slip faults (with up to several kilometres of displacement along them) cut Mesoproterozoic structures in the Fraser Mobile Belt and continue into the southeastern Yilgarn Block for > 50 km (with slight refraction of all structures across the craton margin). Although previously thought to be 1000–1750 Ma old [39], the similarity in orientation and structural relationships to dykes studied in the Albany Mobile Belt suggests a Cambrian age is likely.

Although no dolerite dykes cut Neoproterozoic gneiss of the Leeuwin Complex of the Darling Mobile Belt on the western Yilgarn margin (Fig. 1a), brittle–ductile shear zone arrays (with ubiquitous epidote veining and alteration assemblages in host rocks) of similar orientation and relative structural timing to those in the Albany and Fraser mobile belts suggest that this Early Cambrian deformation event has been pervasive across southern Western Australia and represents crustal extension between Western Australia and East Antarctica (Fig. 12) during subhorizontal ESE–WNW shortening. No geochemical studies have been carried out, limiting any interpretation of magma source and associated tectonic affinities.

Igneous activity in central Australia shows the same relative timing with respect to regional deformation events as documented above for dolerite intrusion in the Albany Mobile Belt. NW-trending fold axes and thrusts in sedimentary rocks of the Officer Basin (pre-dating deposition of the Neoproterozoic Babbagoola Beds [42]) indicates NE–SW shortening prior to extrusion of the Table Hill Volcanics (Fig. 12) in the Early Cambrian or latest Proterozoic (575 ± 40 Ma apparent Rb/Sr mineral age [43]). Deformation may be equated [42,44] with faulting and folding ofthe Proterozoic sediments of central Australia during the Petermann Ranges orogeny at the end of the Proterozoic (end of interval 3 [45], or Early Cambrian [46]). Feeder dykes to the volcanics, as imaged on aeromagnetic data, follow regional NW and NE fault trends [44]. Interpretation of aeromagnetic data [42] shows a NW trend of the form of shallow volcanic horizons, and features interpreted to represent dolerite dykes that strike NW, E and approximately NNW. Normal faults and linear mafic to ultramafic intrusions trending NW are interpreted as indicating a fundamental crustal weakness [42], the displacement on faults and alignment of intrusions again suggesting approximately NE–SW extension.

The Table Hill Volcanics have been considered to be equivalent to the Officer and Kulyong Volcanics of the Officer Basin [47] (?Ordovician ages based on K/Ar dating [48] may reflect isotopic resetting) and the aerially extensive, probably Early Cambrian (based on stratigraphic grounds) Antrim Plateau Volcanics of the Northern Territory ([46] Fig. 12). Volcanic rifts and grabens of Early Cambrian age are also present in the Warburton Basin of northeastern South Australia [49]. The intrusion of dolerite dykes in the Adelaide Fold Belt is bracketed within the period 540–480 Ma [50]. The latter authors consider that, although the dykes intruded during the compressional Delamerian orogeny, the magma was derived from the mantle prior to the onset of this compressional event, possibly at the same time as other Early Cambrian mafic igneous activity throughout Australia.

In the Bunger Hills region of Antarctica (Fig. 12), opposite the Albany Mobile Belt in Gondwanaland reconstructions [8], an E–W trending dolerite dyke has been dated at 550 ± 24 Ma[51]. Further ages of 502 ± 7, 502 ± 12 and 514 ± 6 Ma from other dykes suggests that dyke intrusion occurred during an Early Cambrian event as well as during the Cambro-Ordovician Ross–Delamerian Orogeny, or that there has been isotopic resetting during this younger event. Olivine gabbro and dolerite dykes intruded ~ 1.1–1.2 Ga gneiss and granitoids in the Windmill and Balaena Islands of Antarctica (placed opposite the western Fraser Mobile Belt [8]). A 510 Ma K-Ar (minimum, cooling) age has been obtained for a gabbro from the Balaena Islands [52]. Early Palaeozoic igneous intrusion has also been described in other parts of East Antarctica [53].

The widespread distribution of Early Cambrian volcanics on the Australian platform is unique in the Phanerozoic [49]. The areal extent of terranes with Early Cambrian mafic igneous activity is now even further increased by the earliest Cambrian age for dykes in the Albany Mobile Belt. Dolerite dykes in the Albany Mobile Belt are thought to have been emplaced during the same event that resulted in dyking in the Fraser Mobile Belt, part of East Antarctica and the Adelaide Fold Belt, and extrusion of the Antrim Plateau and Table Hill Volcanics in northern and central Australia. A divergent margin off northwestern Australia between 575 and 510 Ma [54] may be related to rifting of Precambrian terranes (now present in East and Southeast Asia) from Australia [55]. Extrusion of plateau volcanics at province boundaries surrounding the central Australian region has been related to the culmination of deep-seated lithospheric extension [45], although the mechanism for this is not clear.

The exact timing of the change from a divergent margin (extension and sedimentation) to a convergent margin along the proto-Pacific margin of Australia and East Antarctica is uncertain [21]. Marine transgression along the proto-Pacific margin took place in the Early Cambrian and, by the Middle to Late Cambrian, this region had already became a compressional, subduction-related margin resulting in the Ross–Delamerian Orogeny [21,56]. The collision between East and West Gondwanaland also occurred in the Early (–?Middle) Cambrian [18,20,21]. There was, therefore, a probable continuum in the orientation of the regional stress field across a large section of Gondwanaland. The indicated extension direction for mafic volcanism and dyke emplacement is orthogonal to the direction of convergence between East and West Gondwanaland [20]. As noted above, a horizontal orientation of maximum compressive stress is indicated during this interval in the Albany–Fraser Province (and probably the Officer Basin). The continental crust may have acted as a stress guide, transmitting stresses related to the collision between East and West Gondwanaland and initiating extension between cratonic nuclei. The same stress pattern is associated with the Ross–Delamerian Orogeny in the Late Cambrian. Dolerite dyke intrusion continued throughout the Delamerian Orogeny [50]. It is, therefore, important to note that localised extensional terranes resulting in Cambrian volcanism in Australia and Antarctica may be the direct result of convergent tectonics due to, initially, the collision of East and West Gondwanaland and, from the Middle to Late Cambrian, subduction and Andean-style tectonics on the opposite, palaeo-Pacific/Panthalassa margin. Rifting was localised within an overall tectonic regime dominated by a consistent, NW–SE to WNW–ESE horizontal compressive stress. Cambrian extension and volcanism must therefore not be confused (e.g., [57]) with extension related to the Neoproterozoic Gondwanaland rift systems [21] but is, instead, an inherent result of Gondwanaland assembly.

The localisation of rift-bounded basins, dyke swarms and volcanism in Australia during the Early Cambrian is analogous to the localisation of zones of crustal extension (normal-fault-bounded, sediment-filled basins) perpendicular to the collisional front within the Eurasian plate as a result of collisional tectonics in the Himalaya [58]. Deformation due to this collision is also dominated by conjugate strike-slip faulting, extending far into the Eurasian plate. Basaltic volcanism in northeastern China (e.g., Hsing An fissure vol-

canics and plateau basalts) took place within localised extensional zones within an overall compressive tectonic regime. Volcanic fissures and basaltic cinder cones are aligned parallel to normal faults, bisecting conjugate strike-slip faults [58]. The location of rifts is often controlled by proximity to cratonic nuclei.

The extension direction between East Antarctica and the Yilgarn Craton, based on the orientation of dolerite dykes and fault geometries, is not orthogonal to the craton margin. This obliquity, along with the general NNE–SSW to NE–SW extension suggested for the Albany Mobile Belt, Fraser Mobile Belt and Officer Basin during this event is consistent with the model [20] in which a dextral component of displacement is predicted during the Cambrian between Australia and Antarctica.

6. Relationship of younger dykes to rifting on the Western Australian margin

Igneous activity prior to extrusion of the Bunbury basalt during breakup in the Early Cretaceous is rare in the Perth Basin. There is no indication of Carboniferous volcanics or dolerite dykes [61]: the oldest igneous rocks on the western part of the Western Australian margin are Permian phonolite, trachyte and lamprophyre [62], which have been noted in drillcores from the Edel well in the southern Carnarvon Basin [61] (i.e., north of the Perth Basin). Rhyolitic and andesitic conglomerate and tuff of Santonian (Middle Triassic) age or older [63] were penetrated during drilling on the Naturaliste Plateau off the southern Perth Basin (Fig. 1a) (i.e., in the same age bracket as indicated for the Knapp Head dyke examined in this study). NNW- and NW-striking dolerite dykes of the Boyagin dyke swarm along the western Yilgarn Craton margin [64] have not been adequately dated. A 750–550 Ma [65] or Middle Proterozoic (1400–1200 Ma) age [39,64] has been proposed for this dyke swarm. As the Carboniferous Long Point dyke in the Albany Mobile Belt is situated on a direct continuation of a dyke terminating at the southern Yilgarn margin [64], it is possible that Permo-Carboniferous, and possibly Triassic to Middle Jurassic, dykes are also present within the Boyagin swarm.

Carboniferous and younger intrusive ages are of interest as there has previously been little other indication of basic igneous activity in southwestern Australia at these times [54]. The NW- to NNW-strike of these dykes parallels that of Permo-Carboniferous to Middle Triassic normal faults in the Perth Basin (Fig. 1a) formed during the first, NE–SW extensional phase [10]. The fault-bounded Collie, Wilga and Boyup Basins within the Yilgarn Craton (Fig. 1a) also trend NW–SE and preserve down-faulted Middle to Late Carboniferous [59] and Permian glacigene sediments and coal measures [60]. During the latest Jurassic to Early Cretaceous, a change to NW–SE extension (leading to the northwestwards separation of Greater India) occurred along the western Australian margin [10]. A Triassic age for the NNW-striking Knapp Head dyke is favoured over the second possibility for a Cretaceous age based on the Australian APWP (Fig. 10) as it would represent intrusion into an active extension fracture during this event. During Early Cretaceous volcanism, such fractures would strike subparallel to the maximum extension direction, and would not dilate unless magma pressures were great.

Young dykes in the Albany Mobile Belt are, therefore, interpreted as corresponding to the onset and termination of the first intracratonic rifting event during Permo-Carboniferous to Middle Triassic NE–SW extension between Greater India and the Western Australian Shield during sinistral transtension along the western Yilgarn Craton margin [10]. No dykes have been found in the Albany Mobile Belt that intruded during the northwestwards separation of Greater India from Australia in the Early Cretaceous nor during breakup with Antarctica and formation of the Bremer Basin [7,9,10].

The intrusion of dolerite dykes during the first event in the formation of the Perth Basin in Precambrian terranes inboard of the basin may be explained in terms of detachment models [66] in which basaltic underplating and dyke intrusion is confined to the upper plate passive margin

which is deformed in a broad flexure above a zone of underplating. The western margin of Western Australia bears many features that are characteristic of a rifted upper plate margin [10], and broad domal uplift and flexuring may have resulted from thermally induced buoyancy and igneous underplating.

7. Conclusions

Three periods of dolerite dyke and sill intrusion have been distinguished in the Albany Mobile Belt. The first, earliest Cambrian phase took place during crustal extension between the Yilgarn and East Antarctic cratons along with the formation of extension fractures bisecting regional ENE- and SE-striking conjugate strike-slip faults and shear joints. This event appears to have been contemporaneous with mafic volcanism in central and northern Australia, and mafic intrusion in East Antarctica. Faulting and dyke intrusion are considered to represent extensional responses to a horizontal maximum compressive stress related to Gondwanaland assembly. It is highly likely that a large number of E–W dykes in the southern Yilgarn Craton previously considered to be Palaeoproterozoic or Mesoproterozoic are of Cambrian age instead.

Carboniferous and Triassic dolerite dykes intruded during NE–SW extension associated with the sinistrally oblique rifting between Greater India and the Western Australian Shield. This event, previously considered to be restricted to the Perth Basin and immediate craton margin, has affected a much larger region within the Albany Mobile Belt and possibly the Yilgarn Craton (which may constitute an upper plate margin?) than previously considered. No mafic igneous activity or tilting related to breakup with Antarctica has been recognised.

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